Financial Risk Management with Derivatives 2 D

Nordea Credit Valuation Adjustment Report

Long Nguyen

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1 Introduction

In modern financial markets, derivatives are commonly traded over-the-counter (OTC), which exposes counterparties to credit risk. Unlike exchange-traded instruments, OTC derivatives are subject to the possibility that one party may default before the contract reaches maturity. This counterparty credit risk can significantly affect the economic value of a derivative and must therefore be incorporated into pricing and risk management frameworks. This project focuses on the computation and interpretation of Credit Valuation Adjustment (CVA) for a foreign exchange (FX) forward contract. CVA represents the expected loss arising from the potential default of a counterparty during the life of a derivative transaction.

In the given scenario, Nordea Bank has entered into a six-year EUR/USD forward contract with Corporate A, under which Corporate A sold EUR 50 million at a forward rate of 1.05. After one year since the start of the trade, there are five years left. Due to changes in market conditions, the current EUR/USD spot rate is 1.15, which implies that the contract presently has positive mark-to-market value for Nordea and therefore exposes the bank to counterparty credit risk.

To quantify this risk, additional market information is provided, including Corporate A's credit default swap (CDS) spread, the implied volatility of the EUR/USD exchange rate, and the expected recovery rate in the event of default. The analysis assumes zero interest rates, flat term structures, and no correlation between market and credit risk factors, which allows for a simplified but conceptually clear CVA framework.

The objective of this project is to analyze the exposure profile of the FX forward over time and to estimate the corresponding CVA, thereby illustrating how market risk and credit risk jointly determine the fair value of OTC derivatives.

2 Contract Overview

The derivative under consideration is a foreign exchange forward contract entered into between Nordea and Corporate A. At maturity, the payoff of the contract from the perspective of Corporate A is given by

$$F_T = (1.05 - X_T) \times 50{,}000{,}000,$$

where X_T denotes the EUR/USD exchange rate at maturity. Consequently, Corporate A makes a profit when the EUR/USD exchange rate at maturity is below the contractual forward rate of $K = 1.05$ and records a loss when the exchange rate exceeds this level. This implies that Corporate A benefits from a depreciation of the euro against the US dollar and is adversely affected by an appreciation of the euro.

From the perspective of Nordea, which holds the opposite side of the transaction, the payoff at maturity is

$$F_T^{Bank} = (X_T - 1.05) \times 50{,}000{,}000.$$

Therefore, Nordea benefits from an increase in the EUR/USD exchange rate and has losses when the exchange rate declines. Since the current spot rate is above the contractual forward rate, the forward contract presently has positive mark-to-market value for Nordea, giving rise to counterparty credit exposure.

The market and credit inputs used in the analysis are assumed to be constant over time. Corporate A's credit risk is characterized by a credit default swap spread of 250 basis points (2.5%) per annum and an expected recovery rate of $R = 40\%$, which implies a loss given default of 60%. The implied volatility of the EUR/USD exchange rate is taken to be $\sigma = 10\%$. Interest rates r in both currencies are assumed to be zero, and no correlation is assumed between market risk factors and counterparty credit risk. All values and exposures are expressed in US dollars for consistency.

3 CVA of the FX Forward Contract

The CVA for each period between t_1 and t_2 is calculated in the following simplified way:

$$\mathrm{CVA}_{t_1,t_2} = (1-R) \cdot \frac{EE_{t_1} + EE_{t_2}}{2} \cdot P(d)_{t_1,t_2},$$

where $P(d)$ is the default probability of the counterparty between t and T, EE_i is expected (positive) exposure at point i and R is the recovery rate of the counterparty.

The implied probability of default

Assume that in case of default the protection buyer has already paid half of the annual CDS spread, then the formulas how to value the **protection leg** (*protection buyer receives this if default takes place*) and **premium leg** (*protection buyer pays this if no default*) are presented as follows:

$$\begin{aligned} \textbf{Protection leg} &= (1-R) \cdot \text{Notional} - 0.5 \cdot \text{Spread} \cdot \text{Notional}, \\ \textbf{Premium leg} &= \text{Spread} \cdot \text{Notional}. \end{aligned}.$$

The present value of a CDS is zero at inception **(no-arbitrage pricing)**:

$$P(d) \cdot \text{Protection leg} - (1 - P(d)) \cdot \text{Premium leg} = 0,$$

where $P(d)$ is the implied probability of default. Substitute **protection leg** and **premium leg** into above equation, then the implied probability of default $P(d)$ is given by

$$\begin{aligned} P(d) \cdot [(1-R) \cdot \text{Notional} - 0.5 \cdot \text{Spread} \cdot \text{Notional}] &= (1 - P(d)) \cdot \text{Spread} \cdot \text{Notional} \\ \Rightarrow P(d) &= \frac{\text{Spread}}{1 - R + 0.5 \cdot \text{Spread}}. \end{aligned}$$

For the spread 2.5% and the recovery rate of the counterparty $R = 40\%$, the implied probability of default $P(d)$ per year is approximately **4.08%**.

Expected exposure (EE)

At time t, the **expected exposure** (EE) is given by

$$EE_t = \mathbb{E}[\max(V_t, 0)],$$

where V_t is the value of the FX forward. Thus, the expected exposure of the derivative for Nordea is expressed as

$$EE_t = 50{,}000{,}000 \cdot \mathbb{E}\left[(X_t - 1.05)^+\right].$$

Assume that the EUR/USD exchange rate follows a lognormal distribution, and the implied volatility is taken as the best estimate of future volatility. Under these assumptions, the expected positive exposure of the FX forward at a future time t can be represented by the value of a European call option. The option is defined with current spot $S_0 = 1.15$, strike $K = 1.05$, volatility $\sigma = 10\%$ and zero interest

rates r. This equivalence allows the exposure profile of the forward contract to be derived using standard option pricing theory.

The following Black-Scholes model can be used to compute the expected exposure:

$$\begin{cases} d_+ = \dfrac{\ln\left(\dfrac{S_0}{K}\right) + \left(r + \dfrac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \\ d_- = d_+ - \sigma\sqrt{T}, \\ C = S_0\,\mathcal{N}(d_+) - Ke^{-rT}\,\mathcal{N}(d_-), \end{cases}$$

where T is the time to maturity and $\mathcal{N}(z) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{z} e^{-\frac{1}{2}y^2}\,dy$ is the Gaussian integral.

Therefore, the expected exposure (EE) is computed as

$$\begin{aligned} EE_t &= 50{,}000{,}000 \cdot \left(S_0\mathcal{N}(d_+) - Ke^{-rT}\mathcal{N}(d_-)\right) \\ &= 50{,}000{,}000 \cdot \left(S_0\mathcal{N}(d_+) - K\mathcal{N}(d_-)\right), \end{aligned}$$

where the final equality follows from assuming a zero risk-free interest rate.

The expected exposure at each future year is calculated using above European call option model. The Python implementation is reported in the Appendix, while the resulting exposure values are presented in the Table 1.

i (Year)	**EE**
0	5,000,000.00
1	5,541,624.99
2	6,219,131.70
3	6,805,805.75
4	7,324,755.32
5	7,793,834.02

Table 1: Expected Exposure (EE) over time.

Instead of relying on a closed-form option representation, the expected exposure can be computed via Monte Carlo simulation by generating future EUR/USD exchange rate paths under a risk-neutral measure and averaging the positive mark-to-market values of the FX forward across scenarios. This is exactly the same distribution assumed by Black–Scholes — Monte Carlo just samples it numerically.

EUR/USD exchange rate follows a **risk-neutral geometric Brownian motion**:

$$dX_t = \sigma X_t dW_t,$$

where σ is the volatility and W_t is a Brownian motion. Thus, the solution for above geometric Brownian motion is given by

$$X_t = X_0 e^{-\frac{1}{2}\sigma^2 + \sigma\sqrt{t}Z},\ Z \sim \mathcal{N}(0,1).$$

The expected exposure is then approximated by the average of these positive values across all simulated scenarios, leading to the following Monte Carlo estimator:

$$\mathrm{EE}_t \approx \frac{1}{N} \sum_{n=1}^{N} 50{,}000{,}000 \cdot \max\left(X_t^{(n)} - K, 0\right).$$

As $N \to \infty$, this converges to the true expected exposure.

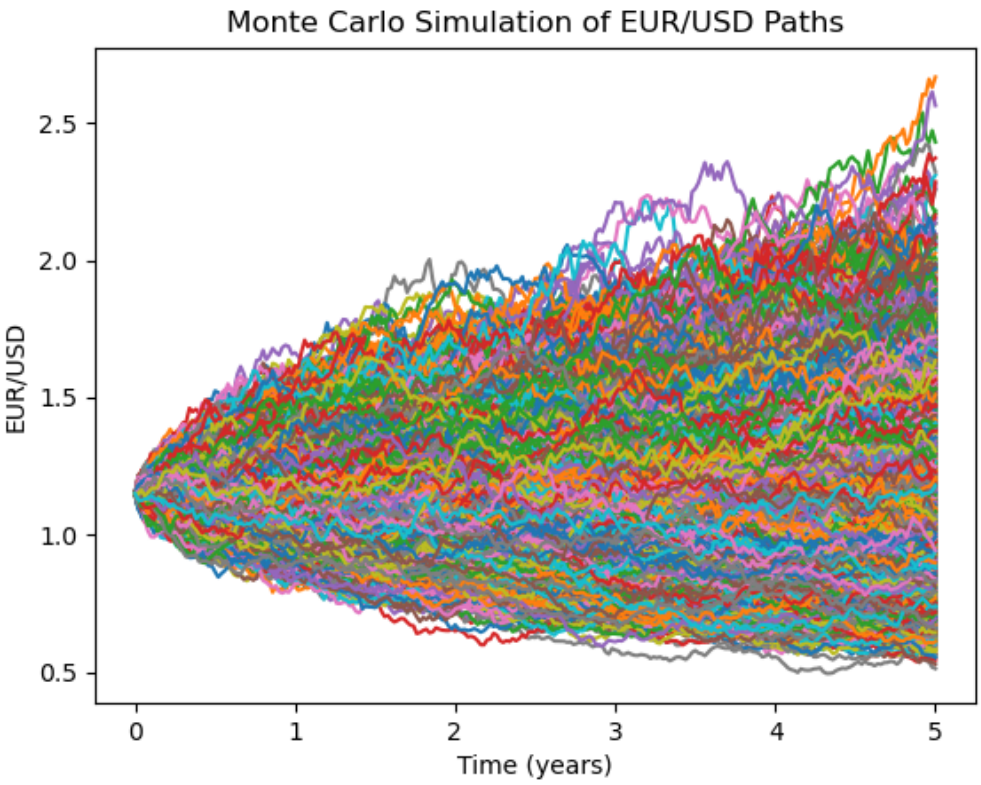

Figure 1: Monte Carlo simulated paths of the EUR/USD exchange rate.

In Figure 1, each path represents a possible future evolution of the exchange rate and is used to compute the future mark-to-market values of the FX forward. Therefore, the final CVA of the derivative for Nordea is given by

$$\mathrm{CVA} = (1 - R) \sum_{i=1}^{5} \frac{EE_{i-1} + EE_i}{2} \cdot P(d)_{i-1,i}.$$

Given the expected exposure (EE), the recovery rate of $R = 40\%$, and the implied default probability of 4.08%, the resulting CVA of the derivative for Nordea is approximately

$$\boxed{790{,}416 \text{ USD}}.$$

The detailed Python implementation used to compute this value is provided in the Appendix.

4 Sensitivity Analysis of CVA

In this section, the sensitivity of the Credit Valuation Adjustment is analyzed with respect to key market and credit risk parameters. Specifically, the impact of changes in the counterparty's CDS spread, the implied volatility of the EUR/USD exchange rate, and the spot exchange rate level on the CVA is examined. This sensitivity analysis provides insight into the primary drivers of CVA and highlights how variations in credit conditions and market factors affect counterparty risk.

4.1 Sensitivity to CDS Spread

Assume that the CDS spread of Corporate A **increases by one basis point**. In the CVA framework, the CDS spread enters the valuation through the implied probability of default, while the expected exposure remains independent of credit risk and is therefore unaffected by changes in the CDS level.

The original CDS spread is 250 basis points (2.5%). After a one basis point increase, the spread becomes 251 basis points (2.51%). Thus, the updated probability of default is

$$P(d)' = \frac{0.0251}{1 - 0.4 + 0.5 \cdot 0.0251} = 0.04098.$$

Therefore, the change in the probability of default is given by

$$\Delta P = P(d)' - P(d) = 0.00018.$$

Since CVA depends linearly on the probability of default $P(d)$, then the following condition holds:

$$\frac{\Delta \text{CVA}}{\text{CVA}} = \frac{\Delta P}{P}.$$

Rearranging the terms yields:

$$\Delta \text{CVA} = 790,416 \cdot \frac{0.00018}{0.0408} \approx 3{,}487 \text{ USD}.$$

Therefore, a one basis point increase in the CDS spread leads to an increase in CVA of about **3,487 USD**, reflecting the higher perceived credit risk of the counterparty.

4.2 Sensitivity to EUR/USD Implied Volatility

To analyze the sensitivity of the CVA with respect to implied volatility, a **one percentage point increase** in the EUR/USD implied volatility is considered. An increase in volatility raises the value of the corresponding European call option, which in turn increases the expected positive exposure. Under the maintained assumptions, the impact on CVA arises solely through changes in exposure. Conducting the same Black–Scholes calculation with $\sigma = 11\%$ produces the updated expected exposure (EE) in the Table 2 on page 9 (see the implementation in the Appendix).

Applying the same CVA aggregation methodology as in the base case yields a revised CVA of approximately **823,771 USD**. Comparing this value with the original

i (Year)	EE
0	5,000,000.00
1	5,691,920.51
2	6,475,271.93
3	7,139,339.70
4	7,721,599.65
5	8,245,318.36

Table 2: Updated Expected Exposure (EE) over time.

CVA of **790,416 USD** implies an increase of about **33,355 USD**. Consequently, a one percentage point increase in implied EUR/USD volatility leads to an increase in CVA of approximately **33,355 USD**.

4.3 Sensitivity to EUR/USD Spot Rate

To examine the sensitivity of the CVA to movements in the EUR/USD spot exchange rate, a **one-figure increase of 0.01** in the spot level is considered. Since the expected exposure (EE) is computed as

$$EE_t = 50{,}000{,}000 \cdot (S_0\mathcal{N}(d_+) - K\mathcal{N}(d_-)),$$

an increase in the current spot exchange rate raises the value of this option, thereby increasing the expected exposure at all future times. Since CVA is directly proportional to expected exposure under the maintained assumptions, a higher spot rate leads to a higher CVA.

The expected exposure is recomputed using an increased spot rate of $S_0 = 1.16$, while all other parameters are held constant. Therefore, the updated expected exposure (EE) is presented in the following Table 3.

i (Year)	EE
0	5,500,000.00
1	5,962,630.51
2	6,604,990.77
3	7,174,740.63
4	7,683,684.36
5	8,146,189.05

Table 3: Updated Expected Exposure (EE) over time.

Using the updated expected exposure, the CVA is recalculated and found to be approximately **838,419 USD** (see the implementation in the Appendix). Compared with the original CVA of **790,416 USD**, this represents an increase of about **48,003 USD**. Therefore, a one-figure increase in the EUR/USD spot rate leads to a substantial rise in CVA.

5 Hedging of CVA Risk

The CVA of the FX forward depends mainly on three factors: the counterparty's default probability (observed from CDS spreads), the EUR/USD exchange rate, and the implied volatility of EUR/USD. Since CVA is the mark-to-market value of counterparty credit risk, changes in any of these variables directly affect the bank's profit and loss. Therefore, the CVA trader aims to hedge all three risks.

5.1 CDS Spread

The sensitivity to CDS spreads comes from the fact that higher spreads imply a higher probability of default and therefore a higher CVA. To hedge this credit risk, the trader buys CDS protection on Corporate A. When CDS spreads widen and CVA increases, the value of the CDS hedge also increases, which offsets the loss.

5.2 EUR/USD Exchange Rate

CVA is also sensitive to movements in the EUR/USD exchange rate because a higher spot rate increases the value of the FX forward and the expected exposure. This risk is hedged by taking offsetting positions in the FX market, typically through spot or forward contracts, so that changes in the exchange rate do not affect CVA.

Accordingly, the trader should take a long position in EUR, or equivalently a short position in USD, in an amount corresponding to the change in CVA. This represents a delta hedging strategy. By holding a long EUR position, the trader benefits from an appreciation of the euro, which compensates for the increase in CVA resulting from a higher EUR/USD exchange rate.

5.3 EUR/USD Implied Volatilty

In addition, CVA increases when implied volatility rises, since higher volatility leads to higher expected exposure. To hedge this effect, the trader uses FX options with opposite volatility exposure, reducing the impact of volatility changes on CVA. In order to neutralize this volatility risk, the CVA trader must construct a hedge whose value increases when volatility rises.

One possible hedging instrument is a **EUR/USD Volatility Swap**. A volatility swap is a derivative contract whose payoff depends on the difference between the realised volatility over the life of the contract and a pre-agreed volatility strike. The payoff at maturity is given by

$$\text{Payoff}_{vol\ swap} = (\sigma_{realised} - \sigma_{strike}) \cdot \text{Notional}.$$

Since the value of a long volatility swap increases when expected or realised volatility increases, it exhibits positive sensitivity to volatility. Holding a long volatility swap therefore provides gains when volatility rises, which can offset the losses caused by an increase in CVA.

An alternative approach consists of hedging the volatility exposure using **EUR/USD Vanilla Options**. Because the expected exposure of the FX forward is equivalent to an option value, CVA carries positive vega κ.

The Vega of a European option is defined as

$$\kappa = \frac{\partial C}{\partial \sigma} = S_t \sqrt{T} \mathcal{N}'(d_+),$$

where $\mathcal{N}'(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}y^2}$ is the density of the standard normal random variable.

This vega is strictly positive for both call and put options. Therefore, by entering a long straddle position—consisting of a long call and a long put with the same strike and maturity—the trader obtains a portfolio with positive vega exposure:

$$\kappa_{straddle} = \kappa_{call} + \kappa_{put} > 0.$$

As implied volatility increases, the value of the option portfolio rises, generating profits that compensate for the adverse CVA revaluation. Therefore, the CVA trader can significantly reduce or eliminate the sensitivity of CVA to changes in EUR/USD implied volatility.

By hedging credit risk with CDS, spot risk with FX instruments, and volatility risk with FX options, the CVA trader can largely eliminate the sensitivity of CVA to these market variables.

6 Hedge Rebalancing After Spot Rate Movement

When EUR/USD spot rate drops to 1.05, the FX forward becomes at-the-money and the expected positive exposure decreases significantly. As a result, the CVA of the transaction is reduced. The hedges previously established by the CVA trader were calibrated for a higher exposure level and therefore become excessive. The trader should consequently rebalance the hedging portfolio by reducing the FX delta hedge, decreasing the volatility hedge, and unwinding part of the CDS protection. CVA hedging is dynamic, and hedges must be continuously adjusted to reflect changes in the expected exposure of the derivative.

7 Market and Economic Discussion

7.1 Market Impact of Similar Client Positions

If many banks hold similar FX forward positions with their clients in large notional amounts, their CVA desks will face very similar risk profiles. In particular, they will all be exposed to the same market risk drivers such as EUR/USD spot movements, implied volatility changes, and credit spread movements of comparable counterparties. As a result, these banks are likely to hedge their CVA risk in the same direction at the same time.

For example, an increase in market volatility would cause CVA to rise across many institutions simultaneously, which prompts widespread demand for volatility hedges and FX options. This collective hedging activity can amplify market movements, which pushes option prices and implied volatility even higher. Similarly, deterioration in credit quality may lead to strong demand for CDS protection, which widens credit spreads further. This feedback loop between CVA revaluation and hedging activity is known as **the vicious circle of CVA**, where market stress increases CVA, which then triggers hedging flows that further increase market stress.

7.2 Corporate Business Model and CVA

If Corporate A is an American corporation exporting goods to Europe, its revenues are naturally denominated in EUR while its expenses are in USD. In this case, the EUR/USD exchange rate directly affects the firm's operating performance and hence its credit quality. When EUR strengthens against USD, Corporate A's EUR revenues translate into higher USD income, improving profitability and reducing the likelihood of default. Conversely, when EUR weakens, revenues decline in USD terms, which worsens the firm's financial condition and increases default risk. Since standard CVA calculations assume independence between exposure and default probability, ignoring this economic linkage would underestimate the true credit risk. Therefore, the presence of such business exposure should lead to a higher CVA charge to reflect the increased likelihood that large exposures occur precisely when the counterparty is more likely to default.

8 Conclusion

In this project, the **Credit Valuation Adjustment** of a EUR/USD FX forward was analyzed using a simplified discrete-time framework. The CVA was computed by combining the expected exposure of the derivative with the counterparty's implied default probability derived from CDS spreads. The exposure was modeled using option pricing theory, which highlights the strong connection between CVA and market variables such as the FX level and implied volatility. Sensitivity analysis demonstrated that CVA is positively affected by increases in credit spreads, volatility, and the EUR/USD exchange rate. Appropriate hedging strategies were discussed, which shows how credit, delta, and vega risks can be mitigated through CDS contracts, FX instruments, and options or volatility products. Finally, broader market implications and economic considerations were examined, which emphasizes that CVA is not only a pricing adjustment but also a dynamic risk management problem with important systemic effects.

Appendix

Imports and model parameters

```
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt

S0 = 1.15
K = 1.05
sigma = 0.1
notional = 50e6
```

Black–Scholes call price function

```
def call_price(S, K, sigma, T):
    if T == 0:
        return max(S - K, 0)
    d1 = (np.log(S/K) + 0.5*sigma**2*T)/(sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    return S*norm.cdf(d1) - K*norm.cdf(d2)
```

Expected exposure computation

```
T = 5
n_paths = 10000
n_steps = 250
dt = T / n_steps

np.random.seed(1)
time = np.linspace(0, T, n_steps + 1)

paths = np.zeros((n_paths, n_steps + 1))
paths[:, 0] = S0

for t in range(1, n_steps + 1):
    Z = np.random.randn(n_paths)
    paths[:, t] = paths[:, t-1] * np.exp(
        -0.5 * sigma**2 * dt + sigma * np.sqrt(dt) * Z)

plt.figure()
for i in range(n_paths):
    plt.plot(time, paths[i])
```

```
plt.xlabel("Time (years)")
plt.ylabel("EUR/USD")
plt.title("Monte Carlo Simulation of EUR/USD Paths")
plt.show()
```

```
EE = []
for t in range(6):
    EE.append(notional * call_price(S0, K, sigma, t))
EE
```

Output:

```
[4999999.9999999935,
 5541624.994005295,
 6219131.698819624,
 6805805.748805915,
 7324755.321226329,
 7793834.02326807]
```

CVA computation

```
LGD = 0.6
pd = 0.0408

CVA = 0
for i in range(1,6):
    avg_EE = 0.5*(EE[i-1] + EE[i])
    CVA += LGD * avg_EE * pd
CVA
```

Output:

```
790415.9872795444
```

Increased volatility ($\sigma = 11\%$)

```
sigma1 = 0.11
EE1 = []
for t in range(6):
    EE1.append(notional * call_price(S0, K, sigma1, t))
EE1
```

Output:

```
[4999999.9999999935,
 5691920.513054355,
 6475271.930553372,
 7139339.696929858,
 7721599.654132794,
 8245318.364150267]
```

Increased spot rate ($S_0 = 1.16$)

```
S1 = 1.16
EE2 = []
for t in range(6):
    EE2.append(notional * call_price(S1, K, sigma, t))
EE2
```

Output:

```
[5499999.9999999935,
 5962630.508755745,
 6604990.768817248,
 7174740.630036686,
 7683684.359583331,
```

```
 8146189.0544122225]
```

CVA computation with increased volatility ($\sigma = 11\%$)

```
CVA1 = 0
for i in range(1,6):
    avg_EE1 = 0.5*(EE1[i-1] + EE1[i])
    CVA1 += LGD * avg_EE1 * pd
CVA1
```

Output:

```
823771.3631107302
```

CVA computation with increased spot rate ($S_0 = 1.16$)

```
CVA2 = 0
for i in range(1,6):
    CVA2 += LGD * 0.5*(EE2[i-1] + EE2[i]) * pd
CVA2
```

Output:

```
838418.9666468904
```